

E-Commerce Database Design Requirements Gathering

For this project, I will be following the given example of an e-commerce database. I will be working on this project as an individual, so I will scale it appropriately. I will be using MySQL as my DBMS and Python for my business logic. The requirements of this specific DB design are the following:

- **Data Requirements: Tables and Attributes**

- Customer
 - The Customer table includes the primary key CustomerId, FirstName, LastName, Email, Phone, and Address
- Staff
 - The Staff table includes the primary key StaffId, FirstName, LastName, Email, Role, and DateHired
- Product
 - The Product table includes the primary key ProductId, Name, Description, Category, Price, Stock, and ManagedBy (references StaffId from Staff)
- CreditCard
 - The CreditCard table includes the primary key CardId, CustomerId (references CustomerId from Customer), CardNumber, CardHolder, DateExpires, and ZipCode
- Purchase
 - The Purchase table includes the primary key PurchaseId, CustomerId (references CustomerId from Customer), ProductId (references ProductId from Product), CardId (references CardId from CreditCard), Quantity, TotalPrice, and DatePurchased

- **User Roles and Use Cases**

- Staff – responsible for adding new products, editing existing products, managing product stock, and have the ability to view purchases
 - Adding Product: Staff can enter the details of a new product and add it to the Product table, StaffId will be stored in the ManagedBy attribute

- Editing Product: Staff can also edit various details about a product if needed
- Inventory Management: If more of a specific product becomes available, staff can add units to the quantity
- Viewing Purchases: Staff can view purchase history from Customers when needed
- Customers - can browse products, make purchases, have multiple cards stored on file, and can view their own purchase history
 - Browsing Products: Customers can view the catalog of current products to purchase from the E-Commerce site
 - Making Purchases: The customer can select a product and quantity to purchase. If the customer has a card on file and the quantity is available, the purchase will succeed
 - Credit Cards: Customers can have zero, one, or multiple cards on file, but must have a card to make a purchase
 - Purchase History: Customers can see a list of their prior purchases